Divvy Station Analysis and Forecasting

CS418 Final Project Report

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1 Introduction

Divvy is Chicago's official bike-share program and consists of over 1800 stations and thousands of bikes. The Streeter Dr & Grand Ave station remains the principal station of analysis because it experiences significant user volumes yet consistently has problems with bike availability. It is located near Navy Pier, a popular tourist destination, which contributes to its heavy use by both locals and tourists.

This project investigates factors influencing Divvy ridership at this specific station during 2023–2024. We study the effects of user behavior, time of day, day of week, seasonality, and weather conditions such as temperature and wind speed. The objective is to understand usage trends, identify operational inefficiencies (such as shortages or dock congestion), and forecast ride demand using statistical and deep learning models.

2 Data Description and Preprocessing

We used 2023 and 2024 Divvy trip data collected from Chicago's open data portal, along with hourly weather data downloaded from the Open-Meteo API.

The dataset was filtered to retain only those rides that started at Streeter Dr & Grand Ave, our target station for analysis. Trips with durations shorter than one minute or longer than six hours were removed, as these were considered outliers or invalid entries.

We extracted features such as hour, month, day of week, and weekend indicator from the trip start timestamps. Season labels were also assigned to each entry based on the trip month.

To further analyze rider behavior, we computed the geographic ride distance using the Haversine formula. Hourly weather data—including temperature, windspeed, and precipitation—was merged with hourly ride features using timestamp alignment in the modeling section.

3 Exploratory Data Analysis

We began by examining which stations were most frequently visited. Streeter Dr & Grand Ave ranked as the top station for both ride starts and ends, justifying its selection for deeper analysis.

Hourly ride distribution showed that Divvy usage peaks during the afternoon hours, especially between 2–3 PM. Member usage was more consistent and aligned with weekday commuting hours (morning and evening), while casual users primarily used the service during afternoons and weekends.

User type analysis confirmed that casual users dominate during weekends and tourist-heavy periods, while members have a strong presence on weekdays, consistent with commuter behavior.

On a monthly scale, usage increased steadily through spring and peaked in summer months (June–August), followed by a decline through fall and winter. This seasonal pattern suggests that weather and outdoor conditions play a significant role in influencing ridership.

Preliminary analysis of weather influence revealed that the number of rides dropped on days with rain or snow. Casual riders were more sensitive to weather changes compared to members, who displayed relatively steady behavior year-round.

We also analyzed net ride flow at the Streeter station by comparing the number of check-outs and check-ins during each hour. The morning hours showed a net outflow (more bikes taken out), while the evenings had a net inflow (more bikes returned), indicating a potential need for rebalancing.

4 Key Visualizations and Hypotheses

1. How does average ride duration vary across different times of day and seasons for member and casual Divvy users, and what does this reveal about their riding behavior?

Hypothesis:
The duration of rides at Divvy Bikes depends on daily times and seasonal conditions while weekday usage shows less variation than weekend usage. Non-members should require more time for rides than members who mainly use Divvy bikes throughout seasons when temperatures are warmer and during weekends.

Why this is an interesting hypothesis:
People of different backgrounds including commuters, those living in the area and visitors use Divvy bikes. Understanding demographic usage patterns improves station management, bike allocation, and user experience enhancement.

Insights:
Member users take shorter rides than casual users. The biggest differences are seen during weekend summer afternoons where casual users ride longer. Members show stable ride durations across time and seasons. In winter, both groups show sharply reduced durations.

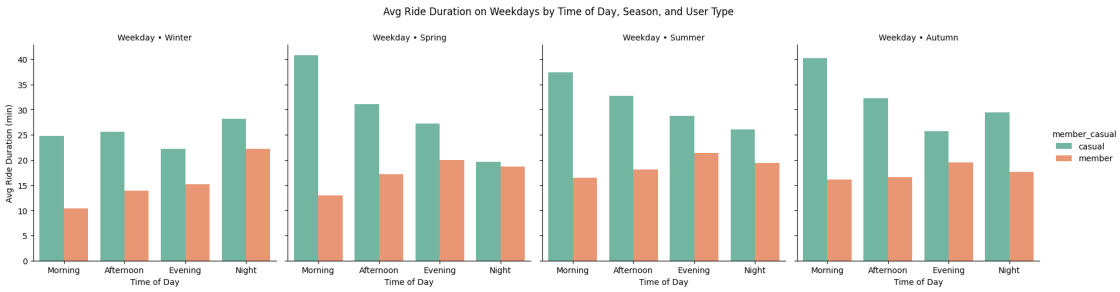

Figure 1: Average Ride Duration by Time of Day, Season, and User Type

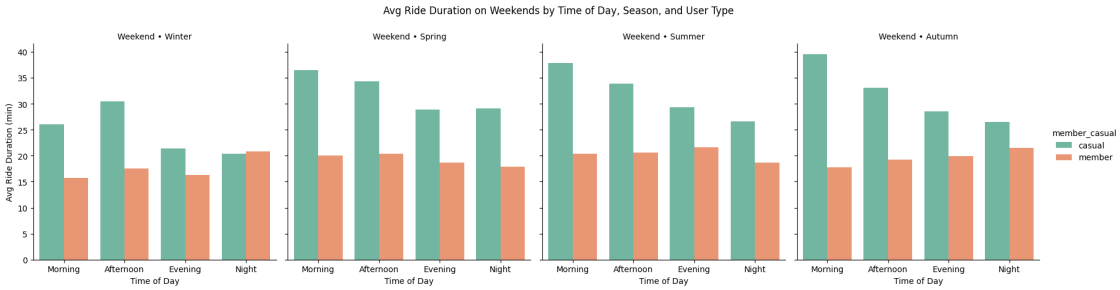

Figure 2: Average Ride Duration by Time of Day, Season, and User Type

2. How does the net ride flow (drop-offs minus pickups) at Streeter Station change across different hours of the day and time blocks, and what does this indicate about potential bike shortages or dock overflows?

Hypothesis:
The net ride flows at Streeter station become different throughout each day, indicating periods of station emptiness (more people picking up bikes) and station fullness (more people dropping off bikes).

Why this is an interesting hypothesis:
Divvy needs to maintain dock availability. Understanding flow imbalance helps schedule bike redistribution and staffing to prevent poor user experiences during rush hours.

Insights:
Negative flow (more pickups) occurs in the morning. Positive flow (more drop-offs) occurs in the evening. Afternoons are more balanced. Nights show low and fluctuating activity. This pattern reflects typical commuter behavior.

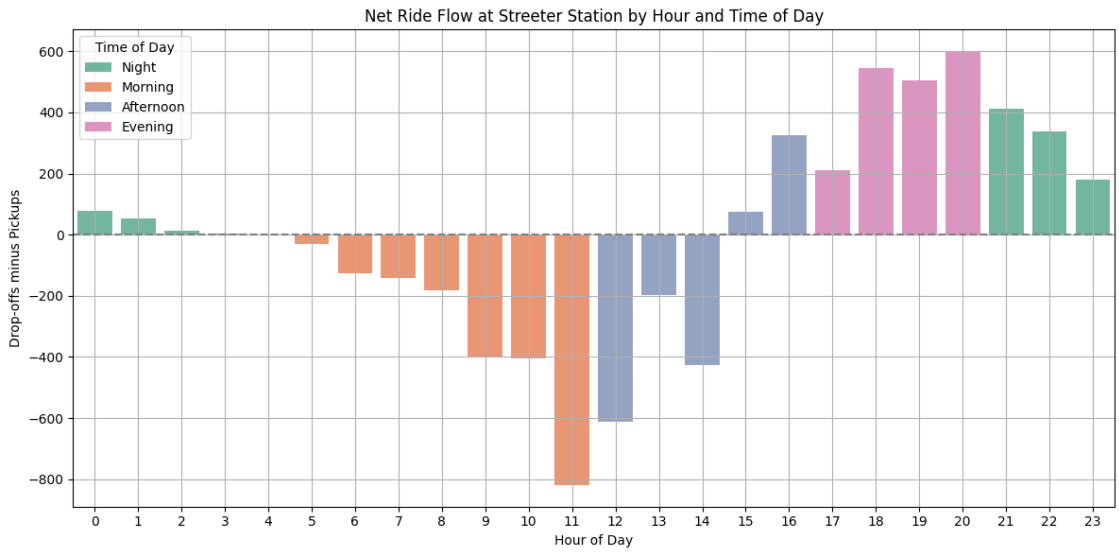

Figure 3: Net Ride Flow at Streeter Station by Hour

3. Average Daily Ride Count By Wind Speed And User Type

Hypothesis:
The number of bicycle rides declines when the wind speed rises and non-members of the bike system experience greater impact than system members.

Why this is an interesting hypothesis:
Casual riders, especially tourists, avoid cycling under high wind conditions which are physically harder.

Insights:
Casual riders show a sharp decline in participation with increasing wind speeds, especially above 8 m/s. Member ride counts remain more stable across wind conditions.

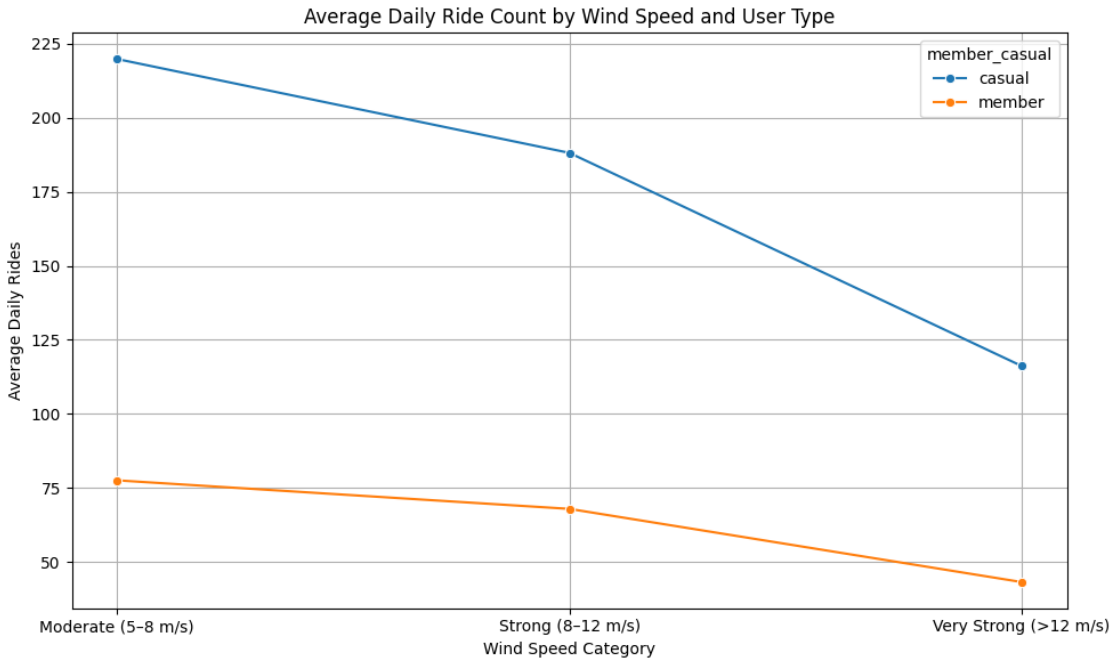

Figure 4: Average Daily Ride Count by Wind Speed and User Type

4. How do different weather conditions and temperature ranges affect average ride volume across weekdays and user types, and what operational insights can Divvy gain from these patterns?

Hypothesis:
Different combinations of weather conditions and temperatures (e.g., clear, rain, snow, cold) influence user type and weekday usage. Casual users reduce activity significantly during weekends with bad weather.

Why this is an interesting hypothesis:
Weather conditions play a major role in rider decisions. Understanding this helps with forecasting demand, planning system maintenance, and issuing weather-based alerts or incentives.

Insights:
Casual riders have peak activity on weekends with clear weather and moderate temperatures. Rainy and snowy days drastically reduce casual rides. Members are more consistent, especially on weekdays.

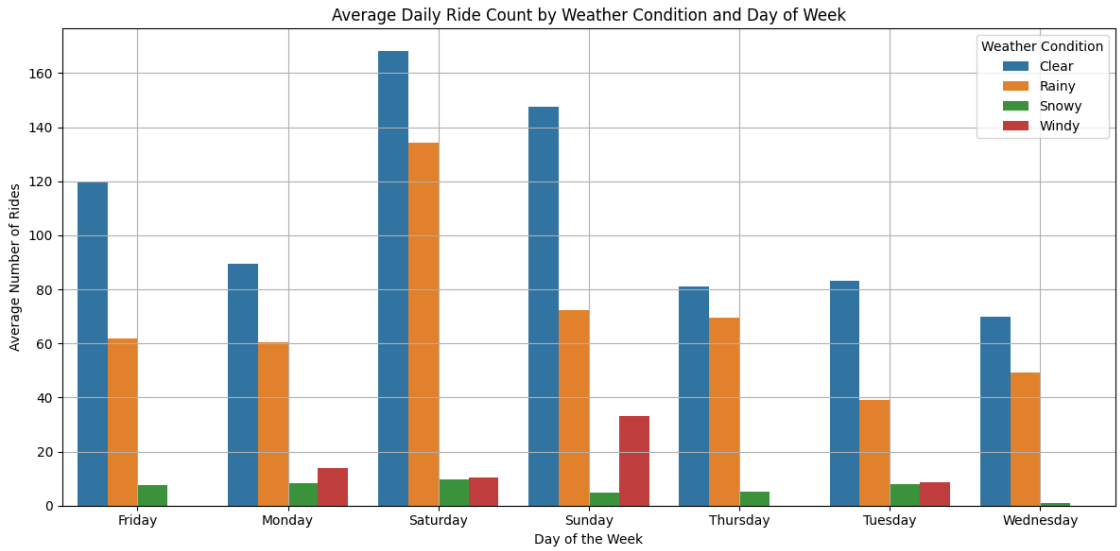

Figure 5: Ride Volume by Weather Condition and Weekday/User Type

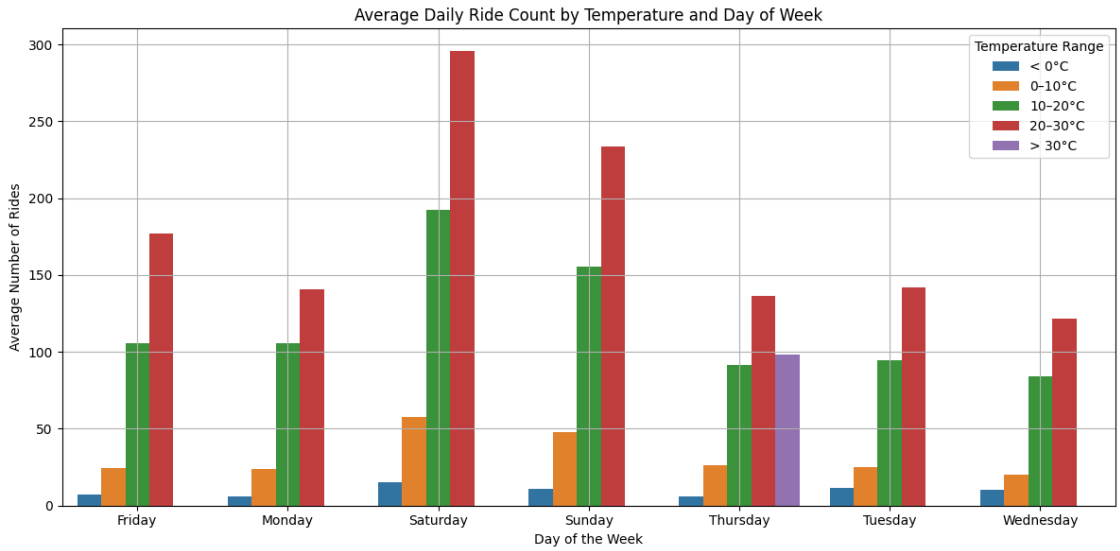

Figure 6: Ride Volume by Weather Condition and Weekday/User Type

5. How does average daily temperature influence total ride volume, and what temperature range represents optimal biking conditions for Divvy users?

Hypothesis:
As the average daily temperature increases, ride volume will increase, peaking within a comfortable range (20–30°C).

Why this is an interesting hypothesis:
Temperature influences demand forecasting, staffing, and promotions.

Insights:
Ride volume is lowest below 10°C, increases between 10–30°C, and drops above 30°C. Casual users show highest temperature sensitivity.

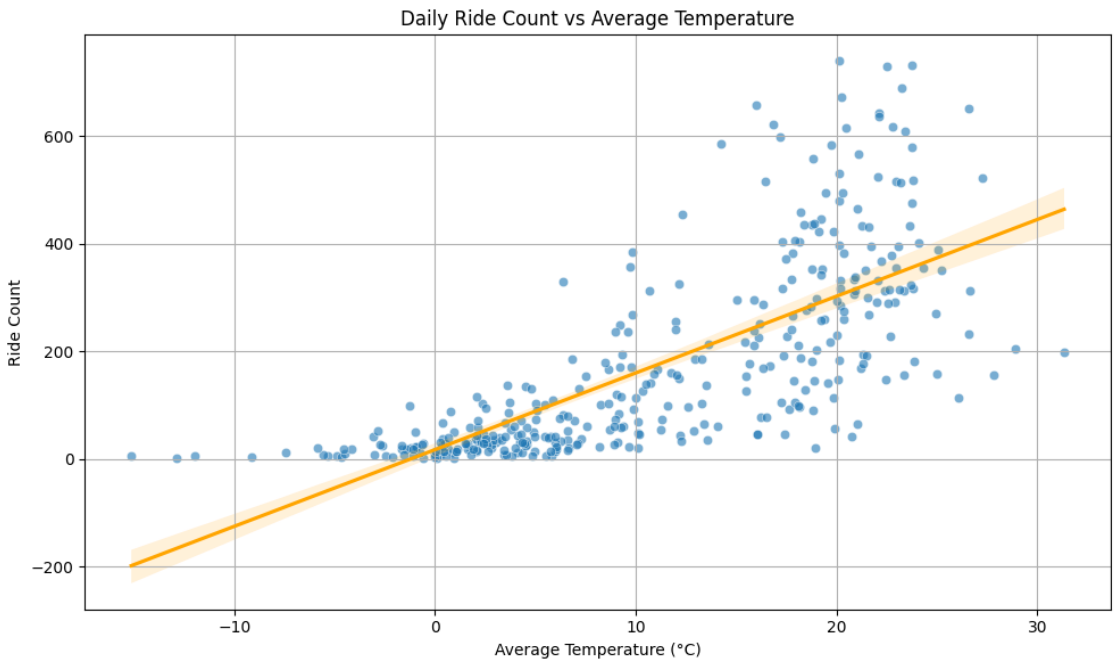

Figure 7: Ride Volume as a Function of Temperature

6. What does the relationship between ride duration and distance reveal about user behavior, and how can it help identify operational anomalies?

Hypothesis:
While most rides show proportional distance-duration trends, some — especially casual rides — will have long durations but short distances, indicating inefficiencies or misuse.

Why this is an interesting hypothesis:
Such outliers might suggest issues like station return delays, getting lost, or exploring leisurely. Identifying them can improve operations, pricing, and system design.

Insights:
Most rides align diagonally (longer distance → longer duration). Casual users show many outliers with long durations and short distances, unlike members whose rides are more linear.

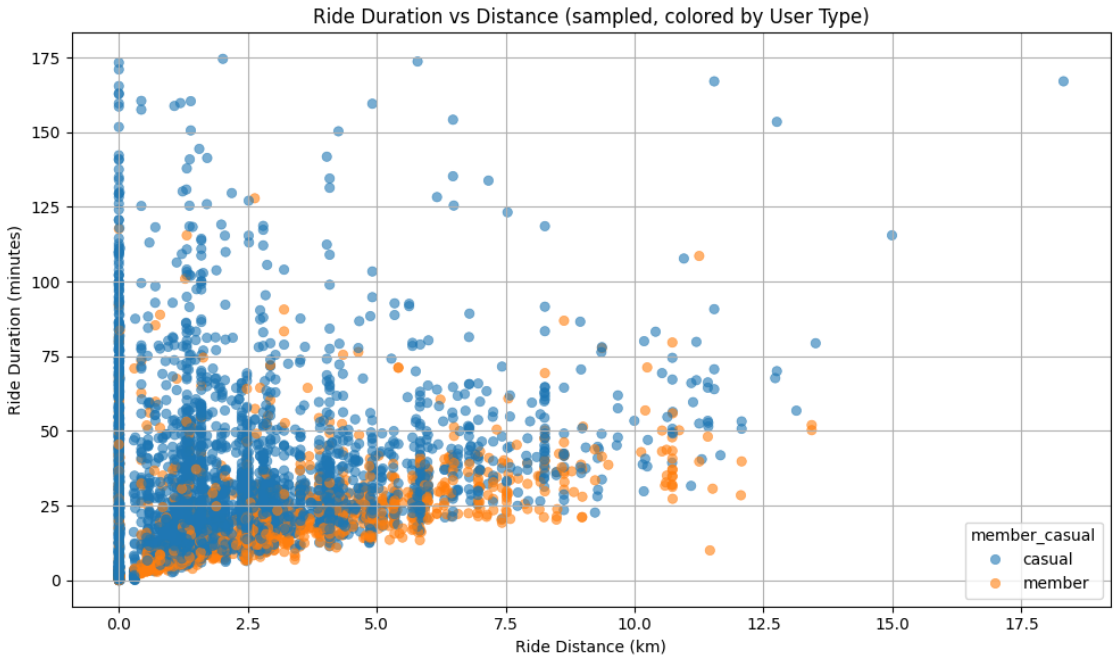

Figure 8: Ride Duration vs Distance (Sampled, Colored by User Type)

5 Statistical and Predictive Analysis

A series of models were explored to understand and forecast demand at the Streeter Dr & Grand Ave station. These models aimed to capture seasonality, temperature trends, and usage fluctuations.

Weighted Moving Average

The weighted moving average model incorporated weather severity (combining precipitation, temperature deviation from 20°C, and wind speed with weights 50%, 30%, and 20% respectively) to smooth daily ride counts. This approach highlighted seasonality patterns more effectively than a simple moving average.

Model Performance:
Mean Squared Error (MSE): 18.89
Mean Absolute Error (MAE): 2.29

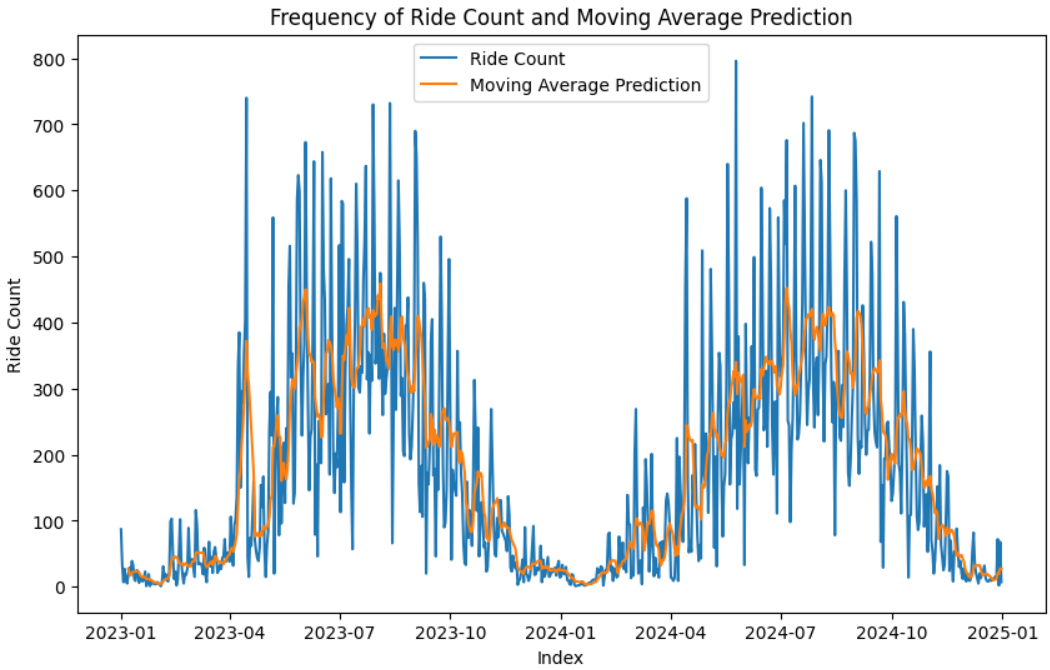

Figure 9: Weighted Moving Average of Ride Count with Weather Severity.

SARIMA Model

A Seasonal AutoRegressive Integrated Moving Average (SARIMA) model was trained using weekly ride data from 2023 to forecast 2024. The model was configured with parameters (1, 1, 1) x (1, 1, 1, 52) to capture yearly seasonality. It successfully modeled long-term trends but slightly underperformed in forecasting holiday-related peaks.

Model Strengths: Captures weekly seasonality and trend effectively, providing interpretable forecasts with confidence bands.

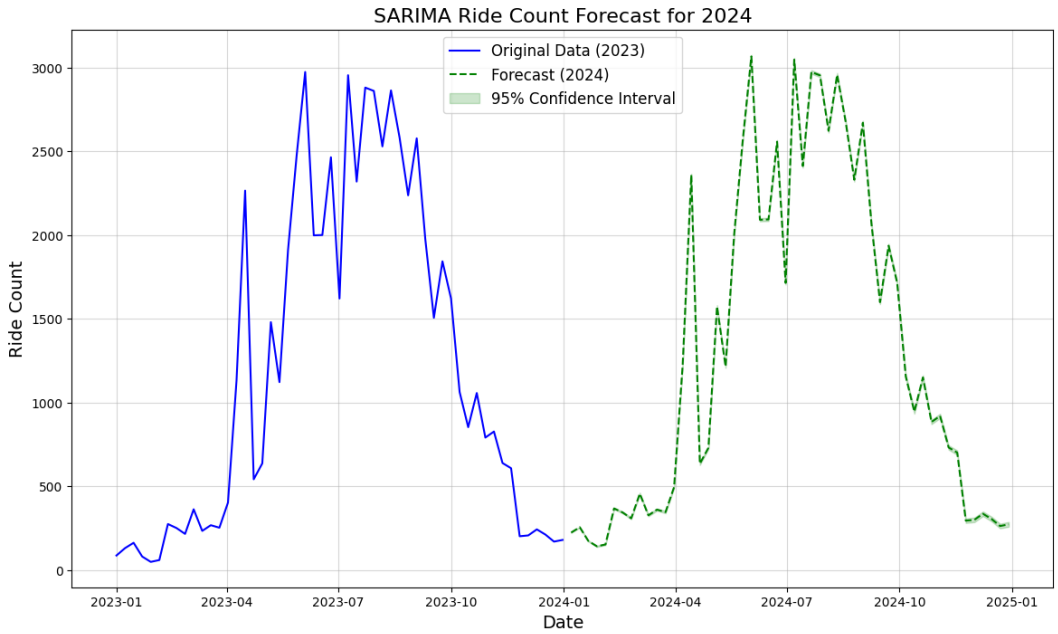

Figure 10: SARIMA Forecast of Weekly Ride Demand.

RNN Model

A Recurrent Neural Network (RNN) was trained on daily ride counts using a 21-day lookback window. The architecture included two recurrent layers and one dense output layer. While the RNN could follow short-term patterns, it underperformed in capturing broader seasonal changes.

Model Performance:
Mean Squared Error (MSE): 12.81
Mean Absolute Error (MAE): 2.09

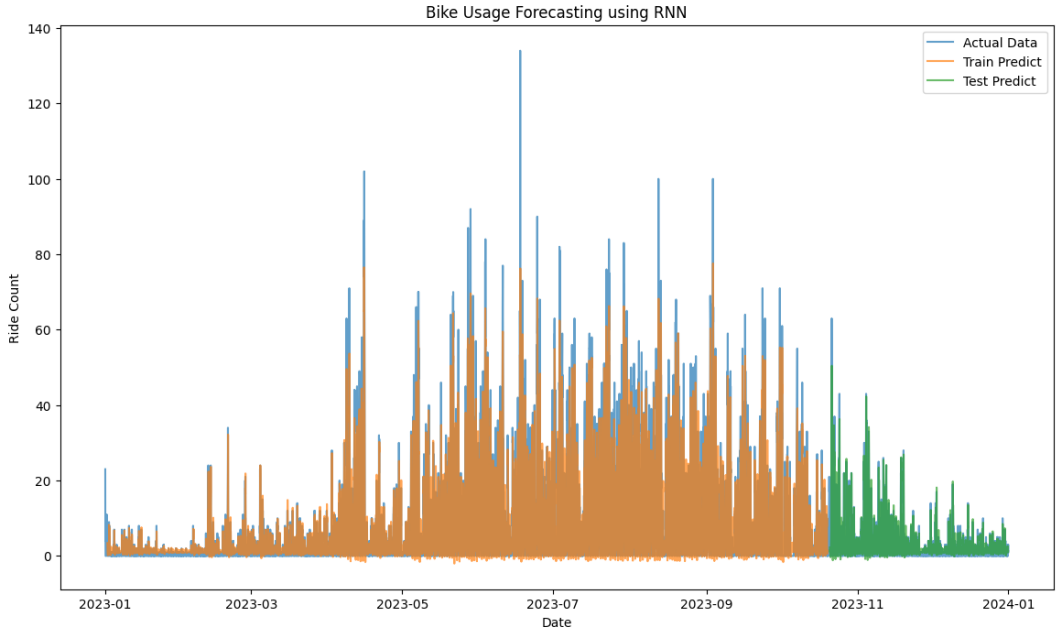

Figure 11: Bike Usage Forecasting using RNN.

LSTM Model

The Long Short-Term Memory (LSTM) model outperformed RNN by capturing longer dependencies and better adapting to seasonal changes. It used a 10-day lookback window and showed strong alignment between actual and predicted ride counts.

Model Performance:

Mean Squared Error (MSE): 13.32
Mean Absolute Error (MAE): 2.04
R^2 Score: 0.599

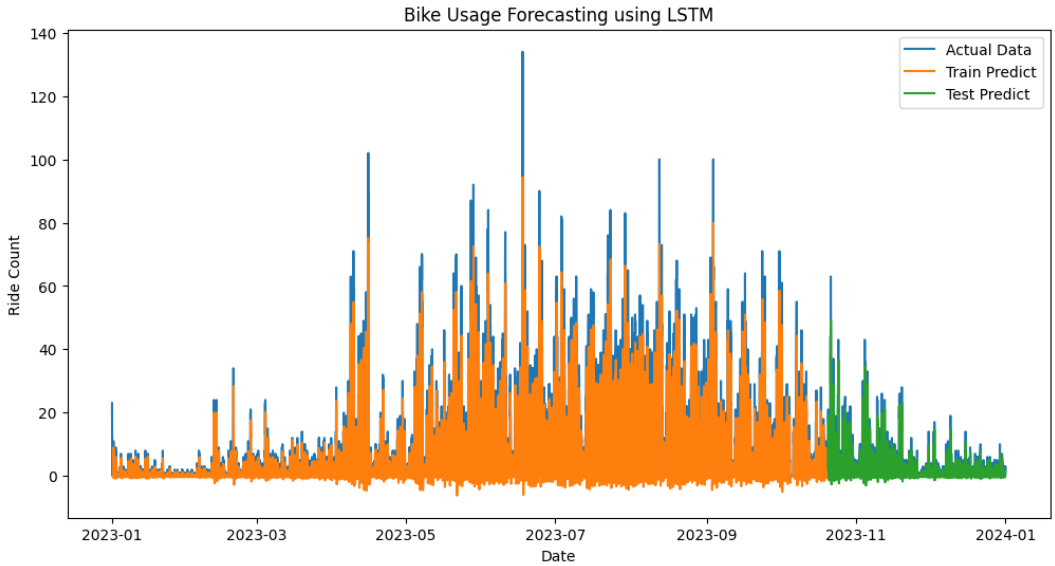

Figure 12: Bike Usage Forecasting using LSTM.

Model Comparison and Inference

Among all models, the LSTM demonstrated the best generalization and temporal learning capabilities. The RNN offered better performance than SARIMA in terms of error metrics, but was less stable on long trends. SARIMA remained the most interpretable, especially for planners focusing on weekly patterns. The weighted moving average served well as a baseline, offering smoothed insights with minimal complexity.

6 Recommendations

Based on our analysis of trip patterns, weather influences, and model forecasts at the Streeter Dr & Grand Ave station, we recommend the following operational strategies:

- **Time-Based Rebalancing:** The station experiences significant net outflow during morning hours and net inflow during the evening. To prevent bike shortages and dock congestion, Divvy should prioritize early morning restocking and post-evening bike redistribution at this location.
- **Weather-Aware Inventory Adjustment:** Since casual ridership drops sharply during adverse conditions like high wind or low temperature, Divvy should dynamically adjust bike supply and staff deployment using short-term weather forecasts.
- **Off-Peak Incentives:** Casual users tend to ride mostly on weekends and afternoons, while members are more consistent across weekdays. Divvy could use pricing strategies (e.g., weekday morning discounts) to shift demand and reduce peak-time pressure.
- **User Education for Inefficient Rides:** Many casual riders take long-duration trips that cover very short distances. This might reflect difficulty finding docks, poor route

planning, or unfamiliarity with the system. Divvy should improve app guidance, station signage, and rider onboarding.

- **Strategic Planning Using Forecasts:** Forecasting models like SARIMA and LSTM have shown the ability to predict seasonal ride demand. Divvy can use these models to plan staffing, bike availability, and maintenance cycles, especially for high-demand summer periods or holidays.

7 Challenges and Future Work

During this project, we encountered several data-related and modeling challenges:

- **Incomplete Weather Data:** Some hourly weather records were missing, particularly snowfall data, which affected the completeness of environmental correlation analysis.
- **Timestamp Alignment:** Trip and weather datasets had different timestamp granularities (hourly vs daily), requiring careful alignment and timezone standardization to ensure accurate merging.
- **Model Limitations:** The SARIMA model underperformed during holiday peaks due to its linear nature. The RNN struggled with capturing long-range temporal trends. The LSTM required tuning and significant computational time in Colab.

For future work, we propose the following improvements:

- Expand the analysis to include **multiple stations** using clustering or PCA to uncover broader network patterns.
- Incorporate **event and public transit data** to understand demand shocks caused by concerts, sports events, or outages.
- Deploy a **real-time dashboard** using tools like Streamlit or Dash to monitor ride demand and weather conditions interactively.
- Experiment with **transformer-based time series models** to capture long-range dependencies more effectively than RNNs or LSTMs.

8 Conclusion

This project investigated ridership trends at the high-traffic Divvy station at Streeter Dr & Grand Ave using 2023–2024 data. We explored temporal and behavioral patterns in usage, analyzed the influence of weather, and built predictive models to forecast ride demand.

Casual riders were found to be more weather-sensitive and leisure-oriented, while members showed commute-driven patterns. Net flow imbalance and seasonal surges highlighted logistical challenges for Divvy operations. Forecasting models such as LSTM showed promising accuracy, while SARIMA offered interpretability.

The findings provide actionable insights to support Divvy's station planning, rebalancing efforts, and user engagement strategies, contributing to a more efficient and responsive bike-sharing ecosystem in Chicago.

Appendix

Visualizations, model evaluation metrics, and source code are available at: https://github.com/shivpatel027/CS418-Project